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PI Name:	Mosier, Kathleen Ph.D.		
Project Title:	Negotiating Crew Autonomy During Space Operations		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	(1) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fischer, Ute Ph.D. (Georgia Institute of Technology) Marquez, Jessica Ph.D. (NASA Ames Research Center)		
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Task Description:	Space-ground collaboration during Artemis and other long missions will continue to be a requirement, given the complexity of space missions and unforeseen events threatening mission safety and success. It is important that crewmembers and mission support on Earth have a common understanding of crew autonomy and operational constraints, and employ a fluid approach to crew autonomy in areas such as the scheduling of tasks and activities. Crew self-scheduling is an integral component of crew autonomy and can serve as a proxy for other autonomy concerns and provide insights that can be applied to other components of autonomy, as it embodies characteristics that may cause friction between space crews and ground, undermine the accuracy of space-ground shared mental models, and negatively impact space-ground collaboration. AIMS of the proposed work are to identify areas of contention in scheduling or where communication concerning scheduling may not be effective, and provide recommendations for how scheduling, as well as other aspects of autonomy could be negotiated and managed in a communication or decision support tool. A review of autonomy issues in space operations, supplemented by selective data on crew autonomy and scheduling from the NASA Human Exploration Research Analog (HERA) and the NASA Scientific International Research In a Unique terrestrial Station (SIRIUS), will deliver: identification of problematic aspects of operational autonomy in space, with particular focus on Artemis; recommendations for how scheduling of tasks and activities can be negotiated, and best practices for managing shifting responsibilities; and definition of required features and parameters for a schedule negotiator technology function with feedback from selected subject matter experts. SIGNIFICANCE of the proposed effort relates to its potential to facilitate collaboration and shared mental models among multi-team system members during Artemis and other long-term space operations. Successful scheduling negotiation is critical to mission success, and solutions will provide insight for other potentially disruptive autonomy elements.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	SIGNIFICANCE of the proposed effort relates to its potential to facilitate collaboration and shared mental models among space/ground multiteam system (MTS) members during Artemis and other long-term autonomous space crew operations. Successful negotiation of task and activity scheduling is critical to mission success and solutions will provide insight for other potentially disruptive autonomy elements. Recommendations for a decision support tool may be applicable for remote autonomous teams on earth.
	Section 5. Task Progress In this effort, scheduling was examined as a probable component of crew autonomy and also as a proxy for concerns around crew autonomy, as it embodies characteristics that may cause friction between space crews and ground support - most notably, different priorities, misunderstandings of priorities, distributed knowledge and expertise concerning constraints and requirements, and potential extra workload for space crews and ground. We used a multi-pronged research strategy, collecting data and information from relevant literature, interviews with analog research participants and NASA autonomy stakeholders, and conducting analyses of analog crew self-scheduling data. This strategy enabled us to combine multiple perspectives from different methodologies and sources. We conducted interviews with NASA autonomy experts/stakeholders, which were transcribed and coded for relevant themes and are included in this report. Our sample of autonomy experts/stakeholders was comprised of people who have experienced or have an interest and a stake in crew autonomy - experts who are qualified to expand on a vision of how things could work under crew autonomy. Their perspectives are rooted in different roles and experiences of autonomy and many interviewees had performed several overlapping roles in research, analog participation, and flight operations. Our questions included interviewee perspectives on: 1. Crew Autonomy – its meaning, components, and motivating factors 2. Crew/Ground Support Relationship – issues and changes in roles during crew autonomy 3. Crew Self-scheduling – 4. Lessons learned from self-scheduling data 5. Requirements for successful implementation of operational autonomy Results Surprisingly, we did not find consistent patterns in our interviews according to stakeholder group – perhaps in part because their multiple roles afforded the interviewees multiple perspectives. Rather, we saw wide variance in expert/stakeholder perspectives with respect to how crew autonomy would be implemented and what it would look like. Predictions ranged from business as usual or minor tweaks to the current system, with much of ground ops looking pretty much the same, to a significantly different concept of operations that requires the crew to possess the expertise, skills, and information necessary to perform a mission independently. Several experts/stakeholders voiced the opinion that preserving the integrity of the space/ground multi-team system (MTS) is going to be more difficult under crew autonomy. Our interviewees suggested that in a crew autonomy concept of operations, ground's role would be more supportive than directive, but ground would continue to be responsible for the 'big picture' and the long view, and would be the overall guardian of mission objectives, goals and priorities, and essential tasks. The issue of who should schedule tasks and activities during autonomy elicited varied responses from our experts/stakeholders. Some interviewees suggested that the crew should self-schedule their tasks and activities; others believed that scheduling should be a shared responsibility between crews and ground. Still other responses built on the theme of ground providing overall guidance and doing some scheduling, but crews having flexibility over their schedules. Benefits and concerns around self-scheduling Several experts/stakeholders noted that crew self-scheduling has the benefit of addressing and implementing crew preferences for activity times – for example, exercising when they like to, having the ability to move operational activities up if they finish another activity early or changing the order in which they accomplish some tasks, or controlling when they perform personal activities. Self-scheduling can be empowering and challenging, and because the crew is 'right there,' they may have situational knowledge to guide activities that ground does not have. One concern that emerged was the tension or tradeoff between control over schedule vs. increase in demands/workload. Our data support the notion that crew autonomy involves crewmembers having some control of their schedule; however, our data did not show a consensus on the scope of crew self-scheduling, in part because of these tradeoffs. Some experts/stakeholders thought the crew should be in charge of the details of tasks and activities as they were closest to the situation. However, although most expert/stakeholder interviewees saw the flexibility and control afforded by

Task Progress:

self-scheduling as a benefit, the idea of crewmembers doing their own scheduling was not regarded as a wholly positive change. Reactions to the notion of crew self-scheduling were often based on the concern that asking the crew to manage all or a large part of their own scheduling would add significantly to their workload and involve extra training, with the rationale that the crew shouldn't be asked to do anything that ground could do for them.

Recommendations for the successful implementation of operational autonomy Integration of relevant literature, self-scheduling data from HERA C6, and input from more than 45 interviews enabled us to pinpoint potential issues, benefits, and concerns from diverse perspectives - as well as ways to address them. In particular, because our interview participants comprised a broad sample across research, analogs, ground operations, and space crew, data from these interviews can serve as a valuable resource for an autonomy concept of operations and provide the basis here for a set of recommendations related to crew autonomy. Notably, the data from HERA C6 and expert insights from interviewees complement and extend findings in extant literature, adding power and validity to our recommendations.

Concepts, definitions, and implementations of crew autonomy As noted, our participants did not share the same conceptualization of crew autonomy, and differences in their notions of autonomy impacted perceptions of aspects of autonomy – such as which tasks and activities should be autonomous, space and ground roles and responsibilities, the length of autonomy periods, how much control space crew should have over their own schedules, or what space-to-ground reporting should be required. This is an important issue to be resolved, as our data suggest that tweaking current operations to enable crew autonomy will not be sufficient for success – more likely is that a crew autonomy concept of operations will require a break from current thinking and current models.

Our data suggest that crew autonomy is not an all-or-none concept; rather, its operationalization must be fluid and its parameters negotiated with ground support, consistent with and conforming to situational affordances and constraints. Crew self-scheduling in some form will likely be a component of any implementation of autonomous operations. Crewmembers respond positively to opportunities to control their own schedules, as evidenced by the high number of spontaneous self-scheduling events in HERA C6. However, the lack of space/ground shared mental models about crew self-scheduling and its relationship to autonomy has led to a gap in understanding how it will function as a basic component of crew autonomy and its implications regarding workload and training requirements.

Recommendation 1. Develop a standard definition of crew autonomy that can be shared by all NASA stakeholders and implemented in autonomy research. This definition will serve as the foundation for future crew autonomy research as well as a crew autonomy concept of operations with parameters for negotiation and navigation of autonomy throughout a mission.

Recommendation 2. Continue and develop new research that examines a consistent implementation of crew autonomy across high-fidelity analogs – that is, analogs that incorporate to the extent possible the isolation, risk, performance pressure, time latencies, and other conditions of long-duration exploration missions. These efforts should incorporate systematic research on self-scheduling and task management in the context of autonomous missions.

Training requirements An autonomy concept of operations will require more up-front and different training than current operations, as autonomous crews must be prepared ahead of time to perform tasks and manage systems independently. Because trust between crews and ground will be critical for autonomy, joint crew/ground pre-mission training will be essential to establish a collaborative relationship. Our expert/stakeholder interviewees suggested that joint training and participation in mission planning can be used to communicate new roles and responsibilities, as well as how performance will be monitored and assessed under autonomy, facilitating trust and a good working relationship between space and ground.

Our data suggest that both specialized training – perhaps a crew of specialists – and cross-training on tasks, automation, and spacecraft systems will be important to facilitate performance of required tasks and activities under crew autonomy. These training requirements represent a shift from heavy reliance on ground for last-minute task assistance to early and thorough preparation to manage tasks and activities in space.

Recommendation 3. Pre-mission joint training is needed to prepare crews and ground for crew autonomy.

This should include: • Training to establish shared understanding of mission priorities and goals, new roles and responsibilities, task/activity capabilities and constraints • Training for new tasks, such as crew self-scheduling, to ensure critical onboard expertise • Training on spacecraft systems and automation, as well as operational and science tasks • Training to maintain space/ground trust to preserve the coherence and integrity of the MTS

Features of enabling technology Technology to enable crew autonomy should be geared toward interrelated aspects of space operations: 1) accomplishing operational tasks and activities, and 2) establishing and maintaining MTS common ground. Space crews will need an archive of readily available information and procedures to help them understand objectives and priorities, make decisions, and perform tasks. As well, supporting technology could help reduce crew workload by automating some tasks and accomplishing reporting functions.

Our data on self-scheduling suggest that a Playbook-type communication and decision support tool could be enhanced to incorporate more information about constraints and potential conflicts, and to make suggestions about the 'best' schedule that fits a given day or situation. This would facilitate the creation of effective schedules and reduce the workload involved.

Functions such as automatic reporting and sharing of activity status updates, currently part of International Space Station (ISS) and analog protocols, have the added benefit of facilitating communication, information sharing, and common ground among members of the MTS. Several interviewees commented on the potential extra workload to maintain MTS situation awareness and shared situation models during autonomy. Many suggested additional ways that input and status information could be shared automatically, using cameras or a Playbook-type communication tool rather than depending on a conversation.

Recommendation 4. Because crews cannot count on timely input and information from ground (given communication delays), technology support for autonomy should include an archive of information and procedures related to task execution that is comprehensive and easily accessible.

Recommendation 5. Technology for crew autonomy should reduce crew workload and support the establishment and maintenance of MTS common ground. This dual goal can be facilitated though shared scheduling software, automation of some operational tasks, and automatic reporting of activity and task status updates.

Bibliography Type:	Description: (Last Updated: 02/11/2026)
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Abstracts for Journals and Proceedings	Mosier K, Fischer U, Marquez J. "Negotiating crew autonomy during space operations." ASPIRE-The Human Factors and Ergonomics Society Annual Meeting, Phoenix, Arizona, September 11, 2024. Abstracts. ASPIRE-The Human Factors and Ergonomics Society Annual Meeting, Phoenix, Arizona, September 11, 2024. , Sep-2024